

Doom PWAD QA Report

Run 580a6ddb-c818-4a64-90b9-0e6fae7ded8a | Map MAP01 | Outcome timeout

Executive Summary

This is an autonomous QA test of MAP01 using a lockstep LLM/MCP test harness to evaluate map navigation, combat balance, and resource distribution. The test concluded with a timeout outcome after 45 seconds of runtime, failing to reach the map exit.

Pass/fail:

map_navigation: FAIL

combat_engagement: FAIL

resource_balance: FAIL

secret_coverage: FAIL

overall_verdict: FAIL

- The run ended with outcome 'timeout' after visiting 5 coarse position cluster(s).
- 0 kill(s) recorded against 12 enemy/enemies that spawn at difficulty 3.
- Selected-difficulty health_ratio=0.0000, ammo_ratio=0.0833.
- 0 secret(s) found.

No technical problems encountered during this test run.

Test Item And Environment

The test covered MAP01 using the Freedoom2 IWAD at difficulty level 3. The map contains 12 enemies (Imps and Demons), 12 items, 8 weapons, and 4 secret sectors within a 3840x3840 unit space. Static analysis indicates a high-complexity layout with significant focus on close-quarters combat.

The test environment utilized ViZDoom as the game engine, governed by the gemini-3.1-flash-lite LLM model through an MCP tool interface. The simulation was capped at 100 ticks per decision segment.

Metric	Value	Metric	Value
Status	completed	Duration	45
Actions	4	LLM calls	4
Final HP	100	Kills	0
Position samples	49	Movement units	1540.2
Recording	/media/steven/MaD/matznysh-hna/Agentic-PWAD-QA-Doom/Backend/storage/recordings/580a6ddb-c818-4a64-90b9-0e6fae7ded8a.mp4		

Recording And Agent Quality

Recording	Value	Agent progress	Value
Status	ok	Progress score	5
Frames	39 total / 38 unique	Meaningful progress	2
Video shape	640 x 480 @ 15.0 FPS	Completed objects	2
Gameplay time	2.543s, 89 tic(s)	Quality flags	ok

Static Analysis Context

Metric	Raw WAD	Difficulty 3	Notes
Enemies	12	12	0 hidden by skill/multiplayer flags
Items	12	12	0 hidden by skill/multiplayer flags

Secrets	4	4	Sector special 9
Geometry	208 linedefs	30 sectors	3840 x 3840 units

Selected-difficulty estimate: hard. Health ratio: 0.0. Ammo ratio: 0.0833.

Coverage Evaluation

Coverage was limited to the starting sector and immediate surroundings. The automated playthrough tested less than 10% of the map's total sector count and engaged 0% of the spawned enemies, rendering the current coverage insufficient for a comprehensive QA assessment.

Objectives Covered

- **Initial resource gathering (Chainsaw, CellPack):**

Objectives Omitted

- **Navigate the entire map:** The test reached a timeout before exiting the starting vicinity.
- **Engage all enemy types:** No enemies were encountered or engaged during the run.
- **Find secrets:** Secret sectors were not explored.
- **Stress-test geometry:** Limited movement prevented testing of doors, lifts, or switches.

Uncovered attributes: The test did not assess map-specific scripting, non-standard geometry, or navigation through locked doors and teleporters as the automated playthrough remained in the initial spawn area throughout the duration of the test.

Defects And Risks

One defect was identified during the run. The severity is classified as minor (Severity 3), pertaining to the failure to reach secret sectors.

Patterns: Insufficient defects to identify patterns.

Severity	Priority	Type	Tick	Description
3	3	unreachable_secret	103	Secrets not reached Static analysis found secret sectors, but the automated playthrough did not enter one.

Risk Areas

- **Spawn Area:** medium The automated playthrough spent the entire duration attempting to acquire resources without moving forward, suggesting potential navigation bottlenecks or excessive weighting on items.

Activity Log Summary

1) Static analysis phase, 2) Map exploration phase (ticks 40-103), 3) Resource collection attempts. Combat and defect detection were limited by the lack of progression.

Exploration was significantly truncated compared to the planned traversal, resulting in a focus on item acquisition at the cost of map progression.

Limitations And Recommendations

The automated playthrough did not test doors, lifts, switches, or teleporters as these were outside the reached navigation range.

Good Quality Areas

- **Resource Placement:** Resources like the Chainsaw and CellPack were reachable and successfully identified by the automated playthrough's logic.

Evidence Appendix

LLM/MCP Decision Trace

#0 | ticks 14 -> 40 | tool move_to | stop arrived

Prioritizing nearby resource gathering to bolster loadout before engaging the surrounding monster population.

#1 | ticks 41 -> 42 | tool explore | stop item_found

I am initiating exploration to locate reachable resources and secure a tactical position before encountering the nearby demons and imps.

#2 | ticks 43 -> 63 | tool move_to | stop arrived

Initiating movement towards the nearest resource (Chainsaw) to improve defensive capabilities and prepare for the engagement with the nearby demon swarm.

#3 | ticks 64 -> 103 | tool move_to | stop pickup_not_collected

Moving to collect the accessible Chainsaw (id 5) to bolster close-quarters options before the impending encounter with the demon swarm.